

# 102010671 | Seeed Studio XIAO ESP32-S3 Plus

SKU: 102010671

## Product Overview

The Seeed Studio XIAO ESP32-S3 Plus is powered by the ESP32-S3's 240MHz Xtensa 32-bit LX7 dual-core processor. It includes WiFi and BLE 5.0, a 2.4GHz rod antenna, and consumes only 14uA in deep sleep mode with lithium battery charging management. The Plus variant offers redesigned castellations and pad layout on the back, providing 9 additional GPIOs for advanced Bluetooth projects via SMD board-to-board soldering.

## Features

**High-Performance MCU:** Dual-core ESP32-S3 at 240MHz, Arduino / MicroPython / ESP-IDF compatible

**Advanced Wireless:** 2.4GHz WiFi + BLE 5.0, 100m+ range with U.FL antenna support

**Efficient Power Management:** Four power modes with 14uA deep sleep, integrated Li-ion charging

**Expanded 20 GPIOs:** 9 extra 1.27mm pitch SMD castellation pins beyond standard XIAO nRF52840

**Universal Compatibility:** Standard backside layout (with aligned battery pads, and debugger pads) for more to-be-released XIAO expansion boards

**Classic XIAO Form Factor:** 21 x 17.8mm design with 2.54mm pin headers, single-sided mounting for wearables and space-constrained projects

**XIAO Accessory Add-ons Compatibility:** Natively compatible with all the XIAO accessory add-ons of expansion, connectivity modules (LoRa, CANBus, RS-485), sensors (2-Mic Array, Vision AI, mmWave Motion, Environmental sensing), and actuators (ePaper display, LED Matrix, relay and more).

**Production-Ready Design with Fusion PCBA Support** As a new addition to the Seeed Studio XIAO family, the XIAO ESP32S3 Plus remains the classic XIAO design (thumb-size, SMD, stamp holes). All these features make it a perfect fit for either space-limited applications, or a production-ready System on Module (SoM) for your PCBA designs. Through our Fusion PCBA Sponsorship program, you can not only enjoy the lowest unit prices for all XIAO modules from our Fusion Open Parts Library, but also fabricate and scale your XIAO-based PCBA designs at low cost (30% off) with FREE Design for Manufacturing consulting services, streamlining your journey from prototype to production.

**Seeed Technology Co.,Ltd.**

9F, Building G3, International E City,  
Zhongshanyuan Road,  
Nanshan, Shenzhen, China

**Contact:**

Tel: +86 0755-80695676  
Web: [www.seeed.cc](http://www.seeed.cc)  
Shop: [www.seeedstudio.com](http://www.seeedstudio.com)

## Product Image



Product image

## Specifications

| Item                                                        | Seeed Studio XIAO ESP32-S3                                                                    | Seeed Studio XIAO ESP32-S3 Plus                                                                                     |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Processor                                                   | ESP32-S3R8 Xtensa LX7 dual-core, 32-bit processor that operates at up to 240 MHz              |                                                                                                                     |
| Wireless                                                    | Complete 2.4GHz Wi-Fi subsystemBluetooth Low Energy 5.0 / Bluetooth Mesh                      |                                                                                                                     |
| Memory                                                      | On-chip 8MB PSRAM & 8MB Flash                                                                 | On-chip 8MB PSRAM & 16MB Flash                                                                                      |
| Interface                                                   | 1x UART1x IIC1x SPI11x GPIOs (PWM)9x ADC1x User LED1x Charge LED1x Reset button1x Boot button | 2x UART1x IIC1x IIS2x SPI18x GPIOs (PWM)9x ADC1x User LED1x Charge LED1x B2B Connector1x Reset button1x Boot button |
| Dimensions                                                  | 21 x 17.8mm                                                                                   |                                                                                                                     |
| Power(Typ.)                                                 | Input voltage (Type-C): 5VInput voltage (BAT): 3.7V                                           |                                                                                                                     |
| Circuit operating Voltage:- Type-C: 5V@19mA- BAT: 3.8V@22mA | Circuit operating Voltage:- Type-C: 5V@28mA- BAT: 3.8V@35mA                                   |                                                                                                                     |
| Charging current:50mA(Fast) / 3.8mA(Trickle)                | Charging current:100mA(Fast) / 0.9mA(Trickle)                                                 |                                                                                                                     |

Seeed Technology Co.,Ltd.

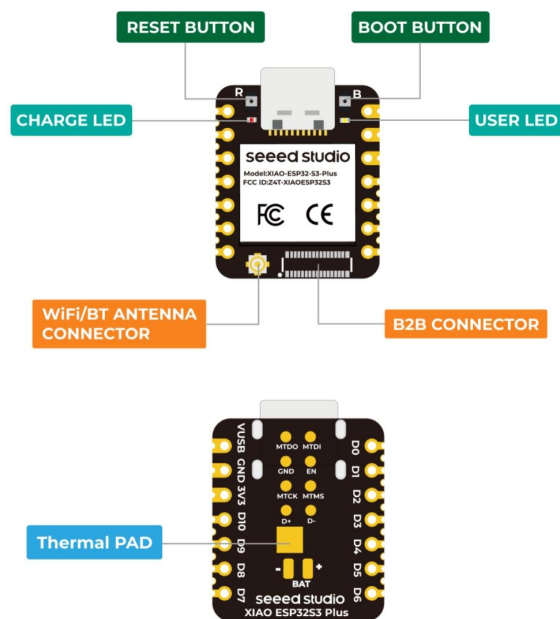
9F, Building G3, International E City,  
Zhongshanyuan Road,  
Nanshan, Shenzhen, China

Contact:

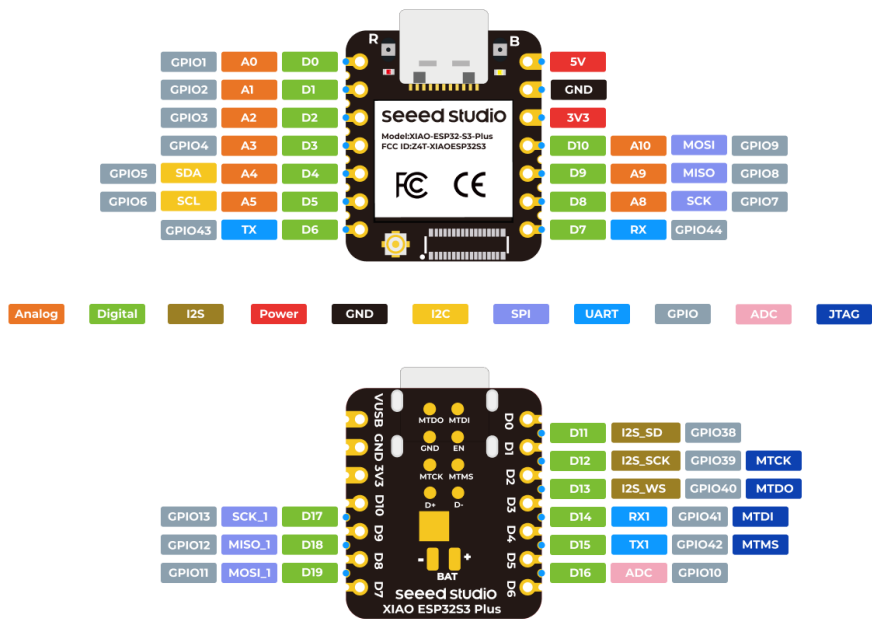
Tel: +86 0755-80695676  
Web: [www.seeed.cc](http://www.seeed.cc)  
Shop: [www.seeedstudio.com](http://www.seeedstudio.com)

| Item                                  | Seeed Studio XIAO ESP32-S3                                                         | Seeed Studio XIAO ESP32-S3 Plus                                                            |
|---------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Low Power Consumption Model(Typ.)     | Modem-sleep Mode: 3.8V/25 mALight-sleep Mode: 3.8V/2 mADeep Sleep Mode: 3.8V/14 uA | Modem-sleep Mode: 3.8V/31.6 mALight-sleep Mode: 3.8V/2.45 mADeep Sleep Mode: 3.8V/33.51 uA |
| Wi-Fi Enabled Power Consumption(Typ.) | Active Mode: 100 mA                                                                | Active Mode: 81 mA                                                                         |
| BLE Enabled Power Consumption(Typ.)   | Active Mode: 85 mA                                                                 | Active Mode: 101 mA                                                                        |
| Working Temperature                   | 40degC 65degC                                                                      |                                                                                            |

## Hardware Overview



Hardware overview image



Hardware overview image